Interior and Exterior Angles of a Triangle

Teacher's Edition

The Period

MA.8.GR.1.5 Solve problems involving the relationships of interior and exterior angles of a triangle.

Must Find a missing interior angle with the Triangle Sum Theorem; find an exterior angle, or a remote interior angle, with the Exterior Angle Theorem; write and solve the equation when the angles are expressions.

Must not Angle sums of polygons (MA.8.GR.1.6); parallel lines and transversals; reflex angles (the angle around the outside is a named error, not an answer).

Target I can find a missing angle of a triangle, inside or outside, with the Triangle Sum and the Exterior Angle Theorem.

MTR **MTR.2.1** — Notes I–II: 180° as torn-off corners, as a sum and as a linear pair · **MTR.4.1** — boards 4 and 7 split; the re-vote · **MTR.5.1** — Notes II: the theorem from the triangle sum and the linear pair together

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up	2–4	6 min
Notes	5–7	10 min
Example 1	8–9	3 min
Your Turn	10–11	3 min
Example 2	12–13	4 min
Your Turn	14–15	3 min
Whiteboards	16–33	12 min
Independent Set	34	6 min
IXL	35	5 min
Total		**53 min**

Say these three things out loud today

1. The three interior angles of a triangle add to 180°.
2. An exterior angle equals the two remote interior angles added — not the angle beside it.
3. With expressions: write the sum or the theorem, solve, put x back, check 180.

Slide by Slide

1. **Interior and Exterior Angles of a Triangle.** Post the learning target.
2. **Warm-Up — On your own. Four minutes. No notes.** Q4 is Notes I before it is said: three angles along a straight line make 180°. *1 retrieves T-B1 — supplementary: the two make 180 (yesterday). 2 retrieves 6.04 — a ladder: the ladder is c, so subtract (last week). 3 retrieves 5.10–11 — a unit rate (last unit). 4 retrieves MA.4.GR.1.3 — angle measures add; three angles along a straight line make 180° (prior grade — the prerequisite this lesson needs).*
3. **Warm-Up — Answers.** Reveal. 1. 97° 2. 8 ft 3. $0.75 4. 70
4. **Vocabulary — The new words, before they are used.** Read each word and its meaning once; the notes use them next.
5. **Notes — Copy the rule and the example.** Inside a triangle, the three make 180.
6. **Notes.** Outside equals the two far ones added.
7. **Notes.** Write the sum, solve, put x back, check 180.
8. **Example 1 — The third angle.** Ask what the three angles of any triangle add to before anyone subtracts: 180.
9. **Example 1 — Worked.** 58 + 71 = 129, and 180 − 129 = 51°.
10. **Your Turn — Same steps, your numbers. Boards up when done.** 39 + 84 = 123, so 180 − 123 = 57°.
11. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
12. **Example 2 — An exterior angle.** Ask which two angles are remote — not touching the exterior angle: the 64° and the 49°.
13. **Example 2 — Worked.** The Exterior Angle Theorem: x = 64 + 49 = 113°.
14. **Your Turn — Same steps, your numbers. Boards up when done.** 72 + 35 = 107°.
15. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).

16–17. Board 1. Two angles of a triangle are 47° and 68°. Find the third angle. → ***65°*** *115 — stopped at the sum [Math Nation names misremembering the sum]; 245 — used 360 [the Triangle Sum is 180]*

18–19. Board 2. An isosceles triangle has two base angles of 52°. Find the third angle. → ***76°*** *128 — used one 52 [the two base angles are equal: both count]; 52 — took all three as equal [isosceles: two equal, not three]*

20–21. Board 3. A triangle's remote interior angles are 38° and 77°. Find the exterior angle x. → ***115°*** *65 — the interior angle beside it, 180 − 115 [Source of Truth, GR.1.5: the exterior angle confused with its neighbour]; 245 — the angle around the outside [Source of Truth, GR.1.5: the reflex angle]*

22–23. Board 4. A triangle's angles are 45°, 60° and 75°. What is the exterior angle beside the 75°? → ***B — 105°*** *(A) equals the adjacent interior angle [the brief's named error: the exterior angle is not the interior angle beside it]; (C) the angle around the outside, 360 − 75 [Source of Truth, GR.1.5: the exterior angle taken as the reflex angle]; (D) only one remote angle [the Exterior Angle Theorem adds both] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

24–25. Board 5. An exterior angle of a triangle is 130°. One remote interior angle is 55°. Find the other remote interior angle. → ***75°*** *185 — added the exterior and the remote angle [the theorem: the remote angles add TO the exterior one]; 50 — the interior angle beside the exterior one [Source of Truth, GR.1.5]*

26–27. Board 6. A triangle's angles are x°, $3x^\circ$ and $(2x+12)^\circ$. Find each angle. → ***28°, 84° and 68°***

28–29. Board 7. An exterior angle measures (5x + 4)°; its remote interior angles measure 52° and (3x)°. Which equation finds x? → ***C — 5x + 4 = 52 + 3x*** *(A) the exterior angle used as an interior angle in the Triangle Sum [Math Nation names writing the equation for the wrong relationship]; (B) one remote angle only [the theorem adds both]; (D) all four angles summed to 180 [the exterior angle is outside the triangle's sum] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

30–31. Board 8. A triangle has angles of 41° and 66°, and its longest side is 9 cm. Find its third angle. → ***73°*** *The the 9-cm side is not needed. 64 — subtracted the 9 as well [the extra number]; 107 — stopped at the sum*

32–33. Board 9. Why is an exterior angle of a triangle always bigger than either of its remote interior angles? Explain. → ***It is the two remote angles added, each more than 0°.*** *'Because it is on the outside' [no reason]; 'it is not always bigger' [the remote angles are each more than 0°]*

34. Independent Set — Six questions. On your own, in writing. Hand out the Independent Set; the same six questions are on the board.

35. IXL — Last five minutes. IXL: every listed skill is required, SmartScore 67.

Variation. Boards 1–2: the third angle, then an isosceles triangle; 3 is an exterior angle; 4 votes on the angle beside the 75°. The pattern breaks at 5 — a remote angle found backwards from the exterior one — then three expressions (6), which equation fits the picture voted (7), a side length that is not an angle (8); 9 is written. The independent set: a third angle, an isosceles triangle's base angles, an exterior angle, a remote angle, three expressions, an equation from the theorem.

Misconceptions to Watch

Board 1. 115 — stopped at the sum [Math Nation names misremembering the sum]

Board 2. 128 — used one 52 [the two base angles are equal: both count]

Board 3. 65 — the interior angle beside it, 180 − 115 [Source of Truth, GR.1.5: the exterior angle confused with its neighbour]

Board 4. equals the adjacent interior angle [the brief's named error: the exterior angle is not the interior angle beside it]

Board 5. 185 — added the exterior and the remote angle [the theorem: the remote angles add TO the exterior one]

Board 7. the exterior angle used as an interior angle in the Triangle Sum [Math Nation names writing the equation for the wrong relationship]

Board 8. 64 — subtracted the 9 as well [the extra number]

Board 9. 'Because it is on the outside' [no reason]

Materials Whiteboards and markers; one paper triangle to tear for Notes I, if there is time. Calculators allowed, printed nowhere.